🔄 The Software Testing Life Cycle (STLC)

From Start to Finish

🏷️ Tags: QA, Process, Methodology

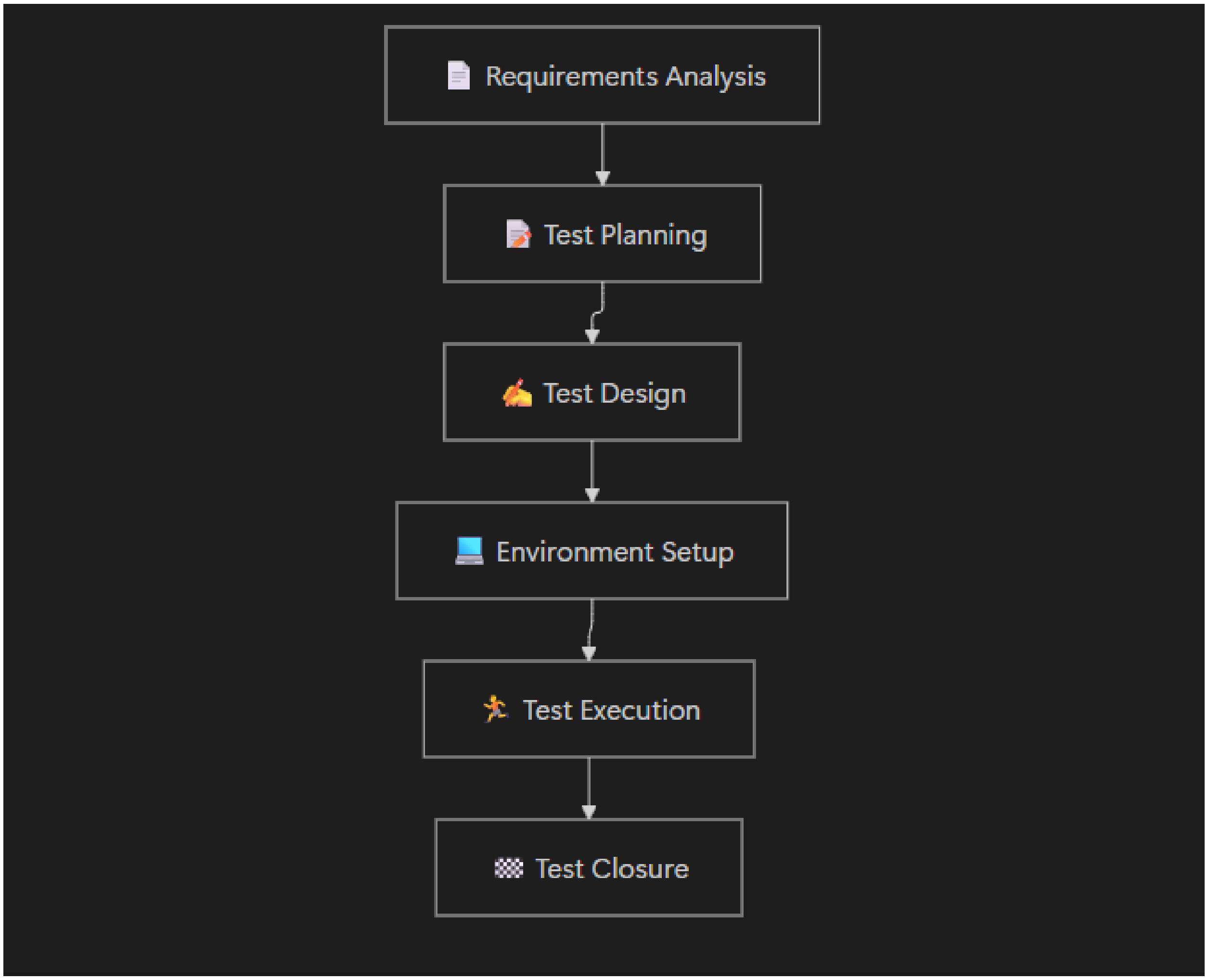
⏱️ Reading Time: 4 min

🚀 TL;DR

Testing isn't just "clicking things at the end."

It is a structured process: **Requirements** ➡️ **Planning** ➡️ **Design** ➡️ **Environment** ➡️ **Execution** ➡️ **Closure.**

📊 The Process Flowchart



🧐 The 6 Phases Explained

1. 📄 Requirement Analysis (The "What")

Goal: Understand what we are building.

We read the documentation (User Stories). We ask the Product Manager questions.

- "Is this button supposed to be blue or red?"
- "What happens if the user has no internet?"

2. 📝 Test Planning (The "How")

Goal: Create the Strategy.

We decide *who* will test, *when* we start, and *what* tools we need.

- Output: The **Test Plan** document.

3. ✍️ Test Design (The "Script")

Goal: Write the instructions.

We write the **Test Cases** and **Test Scenarios**. We prepare the data (e.g., creating dummy users).

- Output: A suite of Test Cases ready to run.

4. 💻 Environment Setup (The "Stage")

Goal: Prepare the server.

We make sure we have a place to test that looks like the real world (Staging Environment). We install the software and configure the database.

- *Note: This can happen parallel to Test Design.*

5. 🏃 Test Execution (The "Action")

Goal: Find the bugs.

This is where we actually test. We follow our Test Cases.

- If it passes: Mark as **Pass**.
- If it fails: Log a **Bug** (This triggers the *Bug Life Cycle*).

6. 🏁 Test Closure (The "Report")

Goal: Summarize the results.

We stop testing. We send a report to the stakeholders saying: "We ran 50 tests, passed 48, and found 2 bugs. The software is safe to release."

🏠 The Construction Analogy

Imagine you are building a house.

1. Requirements: Looking at the architect's blueprints. (Do we want 2 or 3 bathrooms?)

2. Planning: Hiring the workers and buying the tools. (Who does the plumbing?)

3. Design: creating the specific floor plan. (Where exactly does the sink go?)

4. Environment: Clearing the land and setting up the scaffolding.

5. Execution: Actually building the walls and checking if they are straight.

6. Closure: The final walkthrough before handing over the keys.

🧠 Interview Takeaway

If an interviewer asks: *"When do you start testing?"*

Wrong Answer: "When the code is ready."

Right Answer: "I start testing during **Requirement Analysis**. By asking questions early on, I can prevent bugs before a single line of code is written."